

A Second-Order Meta-Analysis on the Effects of Artificial Intelligence Applications on Student Outcomes

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Abstract

Research on AI in education has expanded rapidly, resulting in numerous first-order meta-analyses, yet these studies are limited by inconsistent findings, heterogeneous methods, variable quality, and overlapping primary studies, requiring a second-order meta-analysis to obtain a more reliable estimate of AI's effects. This study examined the overall effect of AI applications on student outcomes, including academic achievement and higher-order thinking skills. We included first-order meta-analyses published between 2020 and 2025, covering primary studies published from 1993 to 2024. Nineteen meta-analyses obtained from electronic databases, involving a total of 58,702 participants, were analyzed. Using a random-effects model, we found a statistically significant moderate mean effect size ($ES = .67$, 95% CI [.55–.78]), indicating that AI technologies meaningfully contribute to student learning. The moderator analysis revealed that the moderators influencing the variability of effects of AI on student outcomes include education level, education field, and publication bias status. Effect sizes were robust across AI types and learning outcome types. Meta-regression showed that sample size and publication year did not predict effect sizes, whereas the number of primary studies did. These findings highlight the need for informed AI

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integration, strengthened pedagogical and institutional capacity, and evidence-based strategies to ensure meaningful improvements in student learning.

Keywords

artificial intelligence, student outcomes, academic achievement, higher-order thinking skills, meta-meta-analysis, improving classroom teaching

Introduction

Advancements in computer-related technologies have made it possible for machines to perform functions resembling human behavior. Since Alan Turing's 1950 paper, "Computing Machinery and Intelligence," and John McCarthy's coining of the term *artificial intelligence* (AI; Haenlein & Kaplan, 2019), AI has been applied across various fields. AI technologies have seen widespread integration within the educational sector, leading to an interest in the field of AI in education (AIEd). Educators and learners in K-12 and university contexts have increasingly used applications and tools driven by AI technologies, such as intelligent robots and adaptive learning system (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020).

Although using AI in educational contexts has been the focus of academic discourse for over three decades (Luckin et al., 2016), recent years have witnessed an increase in research centered on the application of AI for educational objectives. This pursuit has been fueled by the advent of generative AI platforms, such as ChatGPT, which have prompted educators and researchers to explore the integration of AI into classroom practices and research agendas. A plethora of independent studies have been published in AIEd, particularly on the effect of AI tools, platforms, or applications on student outcomes, such as academic achievement or thinking skills (e.g., Feng, 2024; Kumar, 2021; Sung et al., 2019). In this study, student outcomes are conceptualized within two specific domains: academic achievement and higher-order thinking skills. Academic achievement as an indicator of how well students perform academically (York et al., 2015), whereas higher-order thinking skills focus on advanced cognitive processes such as analysis, synthesis, and evaluation, consistent with the levels described in Bloom's taxonomy (Zohar & Dori, 2003). The research on this topic has yielded conflicting results, making it challenging to draw definitive conclusions about the relationship between AI and academic achievement or thinking skills (Dong et al., 2025; Sun & Zhou, 2024). The heterogeneity of study designs, encompassing variations in sample sizes, contexts, and disciplines impedes the ability to derive consistent conclusions from existing research (Dong et al., 2025). An early review of AIEd (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020) revealed a growing interest and impact in AIEd research, yet it also highlighted gaps such as limited integration of deep learning technologies, predominant use of traditional AI methods like natural language processing, and a scarcity of studies that combine AI applications with robust educational theory.

Zheng et al. (2023) offered using the meta-analysis as a method to elucidate the impact of AI on student outcomes. This assertion was due to the limitations observed in

previous reviews of AIED. These limitations included a paucity of research exploring learning achievement and perception, a limited examination of moderator variables, a reliance on qualitative literature reviews that hindered the evaluation of AI effectiveness and moderator impacts, and a lack of clear understanding regarding effective AI implementation in education. Additionally, qualitative literature reviews describe the current state of the research field rather than to estimate effect sizes or examine the moderating roles of variables (Paul & Barari, 2022). Consequently, meta-analyses have been conducted to examine the impact of AI on student outcomes (e.g., Dai et al., 2024; Deng et al., 2025; Hwang, 2022; Setiawan et al., 2025; Wang et al., 2024). These first-order meta-analyses (FOMs) have collectively formed a substantial body of research, providing a comprehensive overview of the current state of knowledge in this field. Despite extensive research, the field lacks a consolidated estimate of AI's impact because existing meta-analyses differ widely in scope, quality, and methodological decisions. An examination of these FOMs reveals conflicting results regarding the impact of AI on student outcomes (e.g., Kim et al., 2017; Yin et al., 2021), which require further analysis for a better portrayal of AI effect on student outcomes.

Conducting a second-order meta-analysis based on these FOMs has the potential to yield numerous advantages. For instance, the effect sizes vary significantly. Some studies report small effect sizes, such as .32, as documented by Yi et al. (2025), while others report substantially larger effect sizes, reaching up to 1.36, as noted by Tlili (2024). This approach affords the ability to calculate a mean effect size based on a substantial sample size. The FOMs concentrate on a single discipline, mathematics (e.g., Hwang, 2022), language (e.g., Liu et al., 2025), or computer science (see García-Martínez et al., 2023). However, some encompass multidisciplinary domains (e.g., Zheng et al., 2023). A second-order meta-analysis has the potential to yield two types of results. Firstly, it can produce shared results that encompass all areas of study. Secondly, it can provide mean effect sizes for common areas of study. Of greater concern is the overlapping problem in FOMs in the literature. FOMs may employ the same independent studies, which may inflate or reduce mean effect sizes. Second-order meta-analysis has been demonstrated to systematically address the issue of overlapping problems. Furthermore, FOM methods are statistically heterogeneous. Some employ Hedge's g or Cohen's d , while others use a random effects model or a fixed effects model. Using a variety of methods has the potential to impede the ability of FOMs to make a reliable inference on the research subject. Additionally, the quality of FOMs exhibits variability, a phenomenon that is examined through the framework of second-order meta-analysis studies. Consequently, there is a necessity for robust quantitative review studies, such as second-order meta-analyses, to provide a more comprehensive depiction of the impact of AI on education, informed by a substantial dataset. Therefore, the objective of this second-order meta-analysis study is threefold: first, to combine the results of the aforementioned FOMs; second, to determine a mean effect size based on a substantial sample; and third, to attain a more profound understanding of the nature of the association between AI and student outcomes through moderator analysis of variables. Accordingly, the present study employs the second-order meta-analysis

method to examine the impact of AI applications used in education on student outcomes. The research questions guiding this study are as follows:

1. Do AI applications used in education have an impact on student outcomes?
 - 1.1. Does the impact of AI applications on student outcomes vary across moderator variables?

The present study focuses on the academic achievements and higher-order thinking skills of the students as student outcomes. The effect sizes were also examined in terms of the moderator variables of AI type (chatbots, AI-enabled assessment tools, multi-type), education level (K12, K12 and higher), education field (language, mathematics, computational sciences, mixed), publication bias level, and quality of the FOMs. A mounting body of research has examined AI tools in education; however, the conceptualization of “AI applications” remains broad and often inconsistent across studies. To enhance conceptual clarity and avoid an overgeneralized treatment of AI, the present study distinguishes among three categories of AI applications reported in the first-order meta-analyses. The following categories are identified: (a) chatbots, (b) AI-enabled assessment tools, and (c) multi-type AI applications that combine or do not clearly separate AI functions. This categorization provides a more focused analytical framework and allows the second-order meta-analysis to meaningfully synthesize heterogeneous AI implementations.

Conceptual Framework

Artificial Intelligence and Its Applications

AI represents the integration of computers, computer-related technologies, machines, and advancements in information communication technology, enabling computers to perform functions that mimic human capabilities (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020). AI aims to develop machines that can think and behave like humans, such as recognizing patterns, reasoning, learning, planning, and anticipating outcomes (Xu et al., 2021). AI uses human-like cognitive processes to solve problems and make decisions (Sajja, 2020). However, the term “artificial intelligence” encompasses various technologies, including algorithms, applications, machine learning, and neural networks, making it challenging to define precisely (Baker et al., 2019). AI is a branch of computer science that aims to understand human cognitive processes and use information systems to replicate their effects (Wang et al., 2024). Consequently, AI involves creating systems that can perform tasks requiring human intelligence, such as learning, reasoning, decision-making, problem-solving, and natural language processing (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020; Xu et al., 2021).

AI has been classified in ways depending on its level of intelligence. Haenlein & Kaplan (2019) distinguish narrow, general and super AI, while Russell & Norvig (2010) differentiate between weak and strong AI. Such classifications illustrate that the development of AI systems progresses across increasing cognitive capability and

autonomy. AI encompasses several core subfields, including natural language processing, planning, expert systems, fuzzy logic, neural and evolutionary computing, computer vision, robotics, and machine learning, each designed to emulate specific dimensions of human cognition (Chowdhary, 2020). Furthermore, AI functions can be categorized by their primary output (Sarker, 2022): Analytical AI (data-driven insights), Functional AI (physical action in robotics/IoT), Interactive AI (communication automation), Textual AI (content creation), and Visual AI (image/video interpretation). Thus AI technology is widely deployed across sectors, including medicine, finance, and trade (Chowdhary, 2020; Xu et al., 2021). Examples include deep learning for COVID-19 detection (Aslan et al., 2021), machine learning-based recommendation systems (Ramzan et al., 2019), fuzzy-logic-driven irrigation automation (Krishnan et al., 2020), and text mining in banking risk analysis (Wei et al., 2019).

One area of AI application is education. Various types of AI have been integrated into educational settings. A review of AI in higher education (Chu et al., 2022) highlighted the primary methods used, including elicitation methods, Bayesian inference and networks, traditional machine learning approaches, deep learning or neural networks, and mixed methods. Another study by Zheng et al. (2023) reviewed articles on AI-enabled education and identified several AI technologies employed in these studies, such as expert systems, natural language processing, and mixed-method approaches. Ouyang & Jiao (2021) proposed AI paradigms in educational practices using techniques such as statistical relational methods, Bayesian networks, natural language processing, Markov decision processes, brain-computer interfaces, machine learning, and deep learning. AI tools have been incorporated into research based on the different techniques used, including intelligent tutoring systems (del Olmo-Muñoz et al., 2022), generative AI (Kumar, 2021), assessment systems (Chen et al., 2024), adaptive learning systems (Hou et al., 2022), and agent systems (Kim et al., 2017). Given the critical role of technology integration in education, different AI tools can provide significant benefits during the learning and teaching process, making this a valuable area for further research.

Artificial Intelligence in Education (AIEd)

The field of AI has made significant strides in simulating human cognition, including learning, reasoning, problem-solving, and perception. Using algorithms and data, AI has demonstrated its capacity to make predictions and decisions, and offer recommendations. These AI-driven tools and platforms are adopted within educational contexts (Dong et al., 2025). The field of AIEd, an interdisciplinary domain integrating computer science, statistics, and education, has witnessed accelerated development and interest. AI has been applied in domains such as teaching, learning, pedagogical innovation, educational data mining, and learning analytics (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020). The rapid advancements in deep learning have had an impact on the field of education, ushering in a new era of personalized learning experiences, transformative teaching methodologies, and significant reduction in teachers' workloads through automation of repetitive tasks (Zheng et al., 2023). Consequently, educators have explored the potential pedagogical opportunities that AI

applications offer to support learners throughout the student life cycle (Zawacki-Richter et al., 2019).

AIED supports tasks such as content creation, automated assessment, and learning analytics; enhances student engagement through human-like interaction (e.g., via chatbots); enables personalized learning tailored to individual needs; and can be used across diverse disciplines including geography, language learning, nursing, mathematics, biology, and computer science (Dong et al., 2025). In language learning, for instance, Generative AI has the capacity to generate realistic dialogue and establish a secure environment where students can hone their language skills. This advantage is especially pertinent for non-native speakers (Lee & Lee, 2024). The potential of AI to expedite and systematize content generation within designated subject domains holds promise for enhancing educational efficiency. This augmentation is predicated on the capacity of AI to alleviate the workload and stress experienced by teachers grappling with shortages and responsibilities (Bozkurt et al., 2023).

Despite the rapid growth in the use of AI in education, several critical gaps remain, particularly in the integration of advanced AI technologies (e.g., deep learning) into classroom settings and the theoretical grounding of research in educational frameworks. Chen, Xie, et al. (2020, Chen, Chen, & Lin, 2020) underscored the fact that while traditional AI techniques, such as natural language processing, are frequently employed in educational contexts, the adoption of more advanced methods remains limited. Furthermore, a significant number of studies in AIED exhibit a weak correlation with educational theories, being driven by computer science or STEM perspectives. In a similar vein, Zawacki-Richter et al. (2019) classified AI applications in higher education into four primary domains: profiling/prediction, assessment, adaptive learning/personalization, and intelligent tutoring systems. However, they also underscored significant gaps concerning ethical concerns, theoretical foundations, and the active involvement of educators in AI design and implementation. This necessity has prompted researchers to examine the impact of AI use on student outcomes, with a particular focus on academic achievement and thinking skills. Research in this domain facilitates comprehension of the nature of AIED and the factors influencing student outcomes.

The Effect of AI on Student Outcomes

A mounting volume of research indicates that AI has the potential to enhance student learning outcomes (Almelweth, 2022; Dong et al., 2025; Lee et al., 2022; Zheng et al., 2023). In this study, student outcomes are used as an umbrella term referring specifically to improvements in achievement and higher-order thinking skills. Academic achievement refers to measurable performance outcomes that reflect the degree to which learners successfully attain instructional objectives within formal educational settings such as schools and universities (Steinmayr et al., 2024). From the perspective of student performance, recent meta-analytic evidence indicates that AI has a generally positive and significant effect on academic achievement. Zheng et al. (2023), synthesizing 24 studies published between 2001 and 2020, found a large effect of AI on

learning achievement and a small effect on learning perception (e.g., motivation, attitudes). A more recent meta-analysis by [Dong et al. \(2025\)](#) reported a strong overall effect size for academic performance across 29 studies, and found that this impact varied depending on factors such as educational level, AI role (e.g., tutor support or intelligent system), intervention duration, instructional strategies, and subject domain. Additionally, a meta-analysis focused on generative AI found a moderate improvement in academic performance among university students, especially in contexts involving text generation, independent learning settings, and smaller sample sizes ([Sun & Zhou, 2024](#)).

These findings suggest that while AI has broadly beneficial effects on achievement, the magnitude of this impact is shaped by the type of AI technology employed (traditional vs. generative) and contextual or instructional design variables. However, contradictory findings have been reported in other studies. [Yin et al. \(2021\)](#) reported the findings of an experimental study, the objective of which was to examine the impact of a chatbot-based micro-learning system on student learning performance. The investigation revealed that there were no statistically significant differences in learning performance between the experimental and control groups. [Hobert et al. \(2023\)](#) conducted a comparative analysis of modes of chatbot use in educational settings. Their findings indicated that the impact of these modes on objective learning outcomes was not statistically significant. Some studies have reported a negative effect. In the context of English as a foreign language, students have reported that they perceive chatbots to be less effective than websites in developing target language skills ([Zhang et al., 2024](#)). The positive impact of AI on students' academic achievement has not been extensively demonstrated or accepted, and there is a paucity of compelling evidence to support this assertion. A comprehensive assessment of these conflicting studies through a scientific approach is necessary to elucidate the impact of AI on college students' academic achievement ([Sun & Zhou, 2024](#)).

Additionally, AI has demonstrated encouraging influence in enhancing students' higher-order thinking skills, particularly their computational and critical thinking skills. *Higher-order thinking skills* describes skills that require learners to critically process, manipulate, and use information, moving beyond mere recall ([McLoughlin & Mynard, 2009](#)). Higher-order thinking is an intentional, self-regulated mental process necessary for tackling complex, ambiguous problems that lack clear answers. It requires the individual to actively construct meaning and structure rather than passively finding a solution ([Resnick, 1987](#)). The literature presents frameworks for categorizing higher-order thinking skills. In response to this variation, [Liu et al. \(2024\)](#) synthesized existing classifications and identified five overarching categories: problem-solving skills, metacognitive skills, critical-thinking skills, teamwork skills, and innovation-development skills. For instance in AI focused studies, a quasi-experimental study by [Huang & Qiao \(2024\)](#) used a STEAM-based AI education model and demonstrated significant enhancements in high school students' computational thinking skills, as well as in their motivation and self-efficacy. In the context of higher education, [Jaboob et al. \(2025\)](#) discovered that the incorporation of generative AI tools improved students' cognitive accomplishment. The study revealed that student behaviors, such as

interaction and self-regulation, amplified this effect. In the context of programming education, the integration of ChatGPT has been observed to exert a substantial influence on students' computational thinking, self-efficacy, and learning motivation (Yilmaz & Karaoglan Yilmaz, 2023). However, Essien et al. (2024) offer a cautionary perspective on the integration of AI in the context of thinking skills. They conducted a study to investigate the impact of ChatGPT on critical thinking skill within a higher education context. The findings indicated that the most substantial enhancements were observed at the lower levels of Bloom's taxonomy. Additionally, they addressed concerns regarding the reliability, accuracy, and potential ethical implications of its implementation in higher education. A number of scholars have expressed concerns that the AIED may potentially compromise the development of deep learning skills and critical thinking abilities, unless it is employed with careful consideration. Consequently, it is imperative that the integration of AI in education be guided by robust pedagogical theories, explicit ethical principles, and designs that deliberately aim to cultivate higher-order thinking skills (Chen, Xie, et al., 2020; Chen, Chen, & Lin, 2020; Dong et al., 2025; Sun & Zhou, 2024; Zawacki-Richter et al., 2019).

Method

Research Design

Synthesizing the findings of first-order meta-analysis (FOM) studies necessitated the second-order meta-analysis method. Second-order meta-analyses are similar to FOM studies. The main difference between these two types of studies is related to the use of data sources. Specifically, second-order meta-analyses employ FOM studies as their primary data, while FOM studies, in turn, use independent studies as their data source (Oh, 2020). Second-order meta-analysis is also referred to as meta-analysis of meta-analyses (Schmidt & Oh, 2013). In this study, meta-analytic processes were conducted according to the PRISMA 2020 stages (Page et al., 2021).

Search Parameters

This section outlines the study's search approach, including the search strategies, inclusion and exclusion criteria, and the screening and selection process, and explains how the studies were systematically identified and selected.

Search Strategies. The Web of Science, Scopus, Eric, Academic Search Ultimate, and ProQuest databases were used to access the data for this study. Searches used the keywords "artificial intelligence", AI, ai, ChatGPT, GPT, Copilot, Gemini, "meta-analysis", "meta-analytic", "systematic review". The "include in title" option was selected during the searches. As the emergence of language model tools has led to a significant shift in the use of AI within educational contexts, the search period between 2020 and 2025 was selected to access current research. A series of inclusion and exclusion criteria were established. The search process was conducted using two main

groups of keywords. The first group included terms related to artificial intelligence, such as “artificial intelligence,” “AI,” “ChatGPT,” “GPT,” “Copilot,” and “Gemini.” The second group consisted of terms, specifically “meta-analysis,” “meta-analytic,” and “systematic review.” We used all possible pairwise combinations of keywords from both groups (for example, AI AND meta-analysis and ChatGPT AND meta-analytic) to form the initial dataset. This method was selected to minimize the risk of overlooking relevant studies; however, it also led to a significant number of duplicate records. Due to the nature of pairing broad AI terms with meta-analytic terms, the initial search yielded 4,558 studies.

Inclusion and Exclusion Criteria. The criteria for inclusion and exclusion in this study are outlined in [Table 1](#).

Screening and Selection. The initial search, which employed two-term keyword combinations across databases, yielded over 4,000 records. The titles and abstracts of the studies yielded by the databases were examined according to the aforementioned criteria. In the first stage of screening, a large number of duplicate studies and papers

Table 1. Inclusion and Exclusion Criteria

Inclusion	Exclusion
FOM studies should be published in English.	Studies published in languages other than English were excluded
The present study incorporates FOM studies published between 2020 and 2025.	Studies before 2020 and after April 2025 were not included
The primary objective should be the identification of AI applications that target cognitive, affective, or behavioral learning outcomes for students.	Studies aiming to predict or model AI-related student behavior or classroom processes in relation to AI use were excluded.
AI applications/tools should be used as instructional technology with a pedagogical approach.	Research that focused on the teaching/education of AI itself has been excluded.
FOM studies should report sufficient and appropriate statistical data to generate generic effect sizes (e.g., Cohen’s <i>d</i> , Hedge’s <i>g</i> , or other effect size types, and statistical indices such as lower limit (LL), upper limit (UP), standard error (SE), and variance value associated with these effect sizes).	Systematic reviews that do not include meta-analyses have been excluded.
FOM studies should be conducted independently to ensure the integrity of the research findings.	In instances where the corrected covered area (CCA) ratio between overlapping FOM studies exceeds 10% (Bracchiglione et al., 2022), FOM studies characterized by a narrow scope and outdated information were excluded.

that did not meet the research criteria were identified due to the overlap across databases ($n = 37$). Two of the 37 initially identified studies were excluded because they were behind paywalls and inaccessible. In accordance with the objectives of the present study and the stipulated inclusion criteria, a total of 35 FOM studies were examined. Sixteen FOM studies were excluded for various reasons, including the impact of factors on AI use ($n = 2$), the examination of knowledge, attitudes, and stress regarding AI ($n = 2$), the prediction of school dropout ($n = 1$), the education of AI ($n = 2$), the comparison of the accuracy rates of AI in answering exam questions prepared by instructors ($n = 3$), and

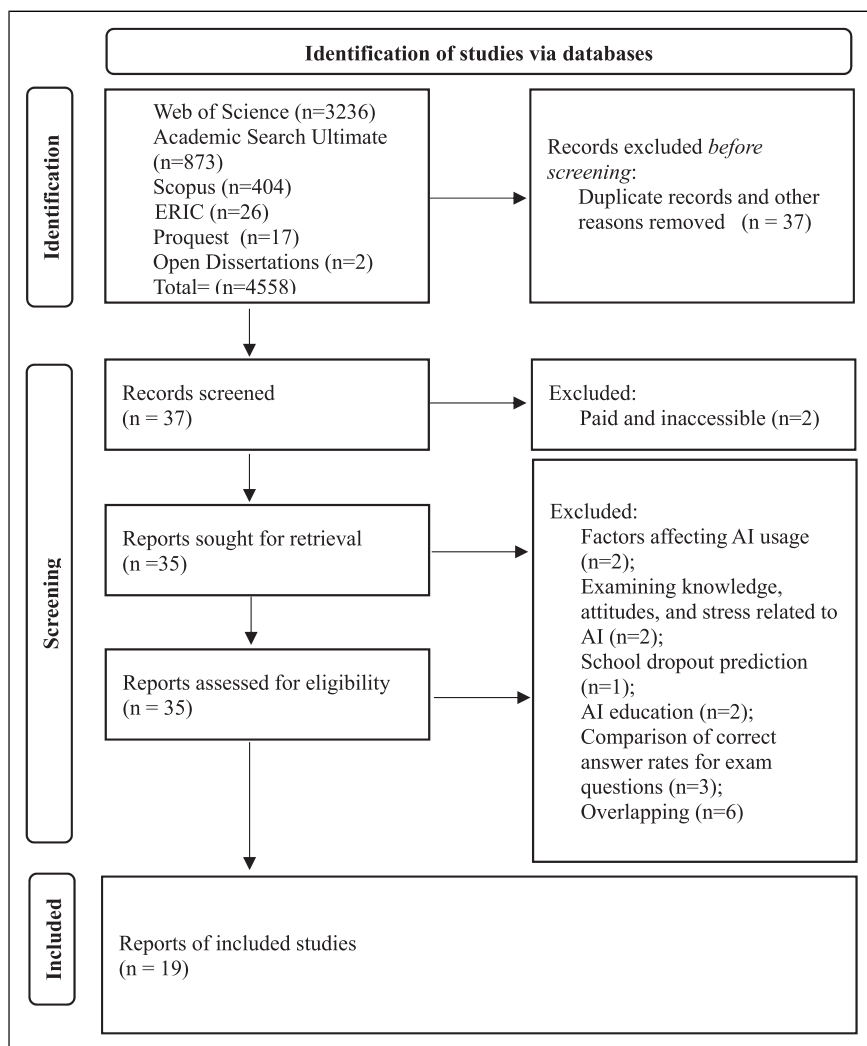


Figure 1. Data flow diagram (Page et al., 2021)

the presence of overlapping data ($n = 6$). Consequently, the dataset for this study encompassed 19 FOM studies. The data flow diagram is presented in [Figure 1](#).

Additionally, the characteristics of the included FOM studies are enumerated in [Table 2](#). FOM studies encompass a minimum of $n = 21$ and a maximum of $n = 85$ independent studies. Of the aforementioned studies, five did not specify the number of participants. According to the studies that did specify the number of participants ($n = 14$), the total number of participants was 58,702. The primary studies encompassed by FOM studies extend from 1993 to 2024. While the $n = 4$ FOM studies are constrained to the K12 level, the $n = 15$ studies encompass both K12 and higher education levels. Although the FOMs included in this study represent diverse educational levels (K–12, higher education, and mixed), combining them is methodologically appropriate in second-order meta-analysis because the unit of analysis is the standardized meta-analytic effect size, not raw primary-study data. First-order meta-analyses already statistically adjust for within-study variability, age differences, instructional contexts, and study designs. Therefore, synthesizing effect sizes across educational levels enables examining broader trends while preserving internal validity, and education level is explicitly included as a moderator to test whether the effects differ across contexts. FOMs emerge from four distinct fields of study, including multidisciplinary studies, mathematics, language studies, and computer science. Among the FOM studies, 13 demonstrate a high level of quality, while 6 exhibit a moderate level of quality. Furthermore, FOM studies concentrate on AI tools ($n = 15$) and chatbots ($n = 4$).

Coding

This section outlines methods for coding and evaluating study features and quality to ensure a systematic assessment of the included meta-analyses.

Coding Study Features. The coding was performed by the first and second authors according to a coding form ([Table 3](#)) that had been prepared in advance. Initially, five FOM studies were coded in conjunction with all authors. Subsequently, the two responsible authors performed the coding independently of each other. A high level of reliability was calculated between the independent codings (Kappa reliability coefficient, $\kappa = .87$). Discrepancies among coders were more prevalent in the AI application category and less prevalent in the student output category and other classifications. The subject of discrepancies among coders was broached. The final decision was made following a consultation with experts in the fields of AI, educational technology, and educational science.

Quality Assessment. The quality assessment was conducted by one of the researchers who also took part in the coding process. The evaluation of FOM studies was conducted by employing the Revised Assessment of Multiple Systematic Reviews (R-AMSTAR) scale, as proposed by [Kung et al. \(2010\)](#). Items 8 C and 8D of the R-AMSTAR scale are clinical practice-oriented. Consequently, these subjects were excluded from the subsequent scoring and evaluation procedures. The quality level of the FM studies was

Table 2. Characteristics of the FOMs

FOM studies	N	ES	LL	UL	Field	Outcome	AI type	Edu.	Bias	Year range	Part.	Quality
Dong et al. (2025)	29	.92	0.61	1.24	Multi-field	A. Ach	Multi-type	K-12 and H	Low	2019-2024	2,657	Medium
Setiawan et al. (2025)	31	.58	0.30	0.85	Multi-field	Combined	Multi-type	K-12 and H	Large	2019-2024	Not	Medium
Dai et al. (2024)	22	.43	0.27	0.59	Multi-field	Combined	Multi-type	K-12 and H	No	2002-2021	4,911	Medium
Dibek et al. (2025)	29	.63	0.46	0.79	Multi-field	HOT	Multi-type	K-12 and H	No	2000-2023	2,966	Medium
Yi et al. (2025)	21	.32	0.20	0.45	Math	Combined	Multi-type	K12	No	2000-2023	Not	High
Hwang (2022)	21	.35	0.22	0.47	Math	A. Ach	Multi-type	K12	No	2000-2022	Not	High
Zheng et al. (2023) ^a	24	.70	0.50	0.90	Multi-field	Combined	Multi-type	K-12 and H	No	2001-2020	2,908	High
Tlili et al. (2025) ^a	85	1.10	0.90	1.30	Multi-field	A. Ach	Multi-type	K-12 and H	Medium	2011-2023	10,469	High
Tlili (2024)	28	1.36	0.95	1.78	Com. Sci	A. Ach	Multi-type	K-12 and H	No	2011-2024	2,765	High
Garcia-Martíne et al. (2023)	25	.72	0.43	1.01	Com. Sci	Combined	Multi-type	K-12 and H	Large	2010-2020	Not	High
Wu (2024)	49	.98	0.75	1.21	Language	Combined	Multi-type	K-12 and H	Low	2013-2024	6,434	High
Liu et al. (2025)	25	.97	0.76	1.18	Language	Combined	Chatbots	K-12 and H	No	2022-2024	4,179	High
Chen et al. (2025)	25	.39	0.01	0.77	Language	A. Ach	AI assessment tools	K12	No	2012-2024	2,523	High
Wu and Li (2024)	21	.65	0.41	0.89	Language	A. Ach	Chatbots	K-12 and H	No	2008-2023	1,232	High
Wang et al. (2024)	45	.70	0.49	0.90	Multi-field	Combined	Multi-type	K-12 and H	No	2010-2022	3,750	High
Zhang et al. (2024)	29	.59	0.35	0.83	Multi-field	Combined	Chatbots	K12	Large	1994-2023	SWD	Medium

(continued)

Table 2. (continued)

FOM studies	N	ES	LL	UL	Field	Outcome	AI type	Edu.	Bias	Year range	Part.	Quality
Lin et al. (2022) ^a	33	.51	0.37	0.65	Multi-field	A. Ach	Multi-type	K-12 and H	No	2012-2021	4,996	Medium
Lee and Lee (2024)	40	.39	0.25	0.53	Language	A. Ach	Multi-type	K-12 and H	No	1993 -2023	8,282	High
Deng et al. (2025)	62	.72	0.57	0.88	Multi-field	Combined	Chatbots	K-12 and H	Low	2023-2024	Not	High

^a= Not reporting a primary study list, Part. = Participant, Not = Not explicitly stated, K-12 and H = K-12 and higher, SWD = Students with Disabilities, Com. Sci = Computational Sciences, A. Ach. = Academic achievement, HOT = Higher-order thinking, Edu. = Education Level.

Table 3. Coding Form

Group	Description/Code
Study reference	Cited in accordance with APA 7.
AI type	If multiple or various AI tools were analyzed together, they were coded as “Multi-type”. Specific AI tools were coded based on how they were reported, such as “Chatbots” and “AI-enabled assessment tools”.
Outcome	Coding was based on the outcome measures synthesized in each meta-analysis. If learning outcomes were evaluated collectively, they were coded as “Combined”. If a single outcome was targeted, it was coded as “Academic achievement” or “Higher-order thinking skills”.
Education level	Education levels covered in the meta-analyses were coded as “K12” and “K12 & Higher”.
Education field	Educational fields examined in the meta-analyses were coded as “Multi”, “Language”, “Math”, and “Computational Sciences”. If more than one field was included, Multi was used; if a specific field was the focus, coding was done accordingly.
Publication bias status	If publication bias was detected statistically, it was coded as “Low”, “Medium”, or “Large”. If not detected, it was coded as “No”.
Publication year range in FOM	Coded based on the publication year range of the primary studies included in the meta-analysis.
Publication year for FOM	Represents the publication year of the meta-analysis.
Participant	Total number of participants included in each meta-analysis.
Quality level for FOM	Quality levels of meta-analyses were coded based on R-AMSTAR scores as “Inadequate”, “Low”, “Medium”, or “High”.
Primary research (n) in FOM	Total number of primary studies included in each meta-analysis.

assessed as inadequate (0 to 11 points), low (12 to 22 points), moderate (23 to 33 points), and high (34 to 44 points; Young, 2017). The quality scores of the FOM studies included in this research ranged from 29 to 36, with an average score of 33.84 and a standard deviation of 2.97. These results indicate that the overall quality of the FOM studies included in the dataset is considered to be high.

Analysis of Overlapping Problem

The overlap analysis was executed using the Excel-based Graphical Representation of Overlap for OVERviews (GROOVE) macro. The Corrected Covered Area (CCA) ratio and color graphical representation analyses were examined according to GROOVE (Bracchiglione et al., 2022). The heat map for CCA is presented in Figure 2. On the other hand, three FOM studies that did not include a list of the basic studies included in the dataset could not be included in the overlap analysis. It was hypothesized that these FOM studies were distinct from other studies and were included in the statistical

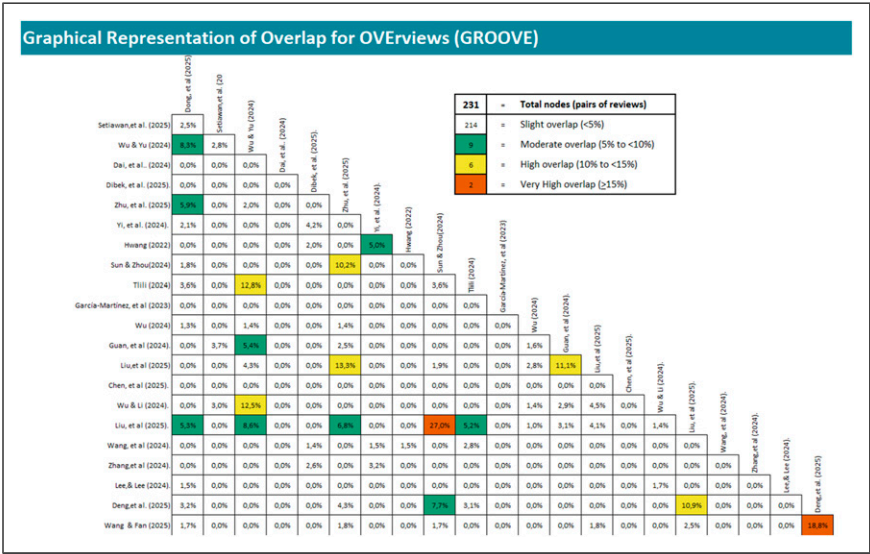


Figure 2. The heat map for CCA of the dataset

analysis. The studies excluded due to the presence of overlap are Wu and Yu (2024), Zhu et al. (2025), Sun and Zhou (2024), Guan et al. (2024), Liu, Guo, He, and Hu (2025), and Wang and Fan (2025).

Analysis Procedures

This section explains the analytical steps employed in the study, covering data analysis, publication bias assessment, and heterogeneity and meta-regression analyses.

Data Analysis. FOM studies exhibit significant variations in terms of their characteristics. The field of FOM studies encompasses a diverse range of participants, AI tools and applications, and a range of research methods. In light of these considerations, a random effects model was employed in the context of meta-analytic statistical analyses (Maitra, 2025). To examine the overall effect of AI applications on student outcomes, the overall effect sizes produced by FOM studies were used as the unit of analysis. In summary, the unit of analysis for this study is at the FOM study level. The statistical analyses were conducted using the CMA.3 program. Moreover, a sensitivity analysis was performed to assess the impact of FOM studies that lacked primary study lists on the overall effect size.

Analysis of Publication Bias. The reliability of meta-analysis results is associated with publication bias (Borenstein et al., 2021). Furthermore, the application of statistical publication bias analyses is not without its own set of advantages and disadvantages.

Publication bias tests are employed to assess the distribution of effect sizes for publication bias (Alexander, 2020). In this study, the Funnel plot, Begg-Mazumdar Test, Egger's Test, and Duval and Tweedie's Trim and Fill tests were performed and interpreted.

Heterogeneity and Meta Regression Analysis. The total amount of heterogeneity present in the dataset was assessed using Cochran's Q statistic (Q), the level of heterogeneity between FMs was assessed using I^2 , and τ^2 statistics were used to interpret the source of heterogeneity (Borenstein et al., 2021). In a similar vein, Q-statistics were used to assess whether the effect sizes between subgroups of categorical moderator variables exhibited variation. Moreover, meta-regression techniques were employed to assess the efficacy of continuous moderator variables in predicting effect size. The meta-regression analysis was executed under the maximum likelihood (ML) model.

Findings

Impact of AI Applications on Student Outcomes

The first research question of the study explored whether AI applications used in education have an impact on student outcomes, and aimed to reveal the mean effect size of this effect. The data set encompasses 19 effect sizes. The effect sizes range from $ES = .32$, representing a small effect size, to $ES = 1.36$, representing a large effect size. The mean effect size was calculated to be $ES = .67$, with a confidence interval (CI) of $[.55-.78]$. In summary, the impact of AI applications on learning outcomes was determined to be moderate. Besides, the total heterogeneity of the dataset was calculated to be $Q(18) = 118.26, p < .001$. The degree of heterogeneity present in the dataset was found to be substantial ($I^2 = 84.78$). Furthermore, it was observed that the heterogeneity of the dataset was significantly influenced not only by sampling error but also by the characteristics of FOM studies ($\tau^2 = .05$).

The distribution of effect sizes according to standard errors was examined using a funnel plot graph (see Figure 3).

A subsequent observation revealed that effect sizes were predominantly located on the right side of the graph. Furthermore, studies with smaller sample sizes were also positioned on the right side of the graph. In this instance, the funnel plot graph is not symmetrical. The Begg-Mazumdar test result (Kendall's $\tau = .32, z = 1.92, p = .06$) did not reveal any publication bias. However, the Egger's Test result indicated the presence of publication bias ($t = 3.33, p = .01$). Conversely, the Duval and Tweedie's Trim and Fill test did not identify the number of effect sizes that would be necessary to ensure the symmetry of the distribution. The results regarding publication bias were mixed. While Duval and Tweedie's Trim and Fill and the Begg-Mazumdar tests did not show evidence of publication bias, Egger's test suggested the possibility of bias, and the Funnel plot also suggested potential signs of bias. Therefore, it can be concluded that there are some indicators of publication bias. Overall, the analyses on publication bias yielded inconsistent results.

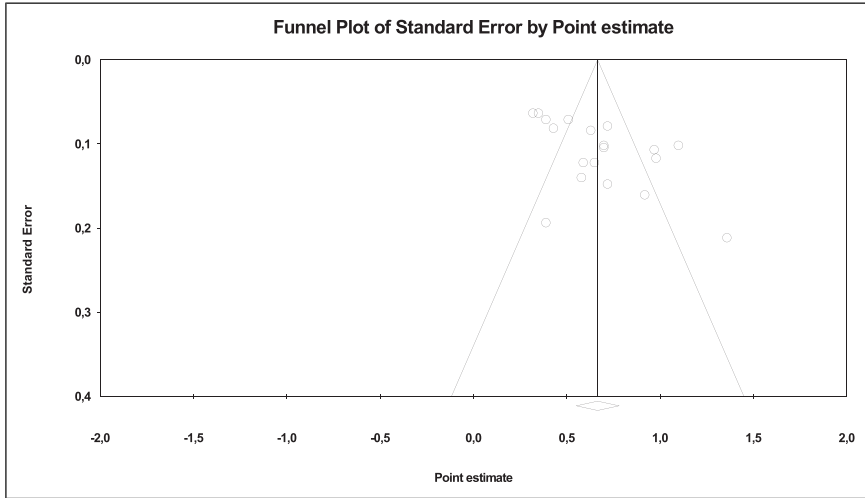


Figure 3. The funnel plot

Moderator Analysis of the Impact of AI Applications on Student Outcomes

The second research question sought to reveal whether the impact of AI applications on student outcomes vary in terms of moderator variables. The moderator analysis examined the mean effect sizes in terms of outcome type, AI type, education level, education field, publication bias level, and quality level of the studies. The results are presented in [Table 4](#).

The effect sizes of FOMs did not differ statistically significantly according to learning outcome types ($Q(2) = .03, p > .05$), AI type ($Q(2) = 1.19, p > .05$), and quality levels ($Q(1) = .62, p > .05$). A statistically significant discrepancy in effect sizes was observed according to educational level ($Q(2) = 6.95, p < .05$). At the K12 level, the effect size was weak ($ES = 0.40, CI = [0.19-0.62]$), while the group encompassing both K12 and higher education produced a moderate effect size ($ES = 0.73, CI = [0.62-0.84]$). Once more, effect sizes vary considerably across educational disciplines ($Q(3) = 8.36, p < .05$). The effect size was found to be weak in the mathematics field ($ES = .34, CI = [.05-.62]$), high in the computational sciences field ($ES = .99, CI = [.62-.65]$), and moderate in the group encompassing language and multi-fields. A similar outcome was observed in the analysis of effect sizes, which demonstrated significant variation depending on the publication bias status of FOM studies ($Q(3) = 8.90, p < .05$). The group with low publication bias yielded high-level effect sizes ($ES = .86, CI = [.62-1.10]$), while FOMs without publication bias produced moderate-level effect sizes. The results of the meta-regression analysis for continuous moderator variables are presented in [Table 5](#).

The number of participants (students) in FOM studies (Coefficient = .00 $p = .69$) and publication years (Coefficient = .10 $p = .07$) do not predict effect sizes in a statistically

Table 4. Moderator Analysis of FOM Dataset

Group	<i>k</i>	<i>ES</i>	<i>LL</i>	<i>UL</i>	<i>Q</i> (bg)	<i>df</i>	<i>p</i>
Outcome					.03	2	.98
Combined	10	.67	.50	.83			
Academic achievement	8	.68	.48	.87			
Higher-order thinking skills	1	.63	.12	1.14			
AI type					1.19	2	.55
Multi-type	14	.66	.53	.79			
Chatbots	4	.74	.49	.98			
AI-enabled assessment tools	1	.39	-.19	.97			
Education level					6.95	1	.01
K12	4	.40	.19	.62			
K-12 and higher	15	.73	.62	.84			
Education field					8.36	3	.04
Multi-field	10	.68	.54	.82			
Language	5	.69	.49	.89			
Mathematics	2	.34	.05	.62			
Computational sciences	2	.99	.62	1.35			
Publication bias level					8.90	3	.03
No	12	.58	.46	.70			
Low	3	.86	.62	1.10			
Medium	1	1.10	.69	1.51			
Large	3	.63	.37	.88			
Quality level					.62	1	.43
High	13	.70	.56	.84			
Medium	6	.60	.39	.81			

significant manner. However, the number of studies included in FOM research (*n*) does predict effect sizes in a statistically significant positive direction (Coefficient = .01, *p* = .01). As the number of independent studies incorporated into FOM research (*n*) rises, the magnitude of effect sizes concomitantly increases.

Sensitivity Analysis Results

A sensitivity analysis was conducted to determine whether the overall effect size was influenced by FOM studies that did not provide a list of primary studies. Table 6 compares the effect sizes of studies that reported their primary study lists (labeled “With list”) with those of the combined dataset that includes both studies that reported lists and those that did not (labeled “With and without list”).

The analysis revealed no statistically significant difference between the two groups (*Q* (1) = .13, *p* = .72). Furthermore, the level of heterogeneity showed only a negligible change between the groups. Therefore, we can conclude that excluding FOM studies without

Covariate	Coefficient	Standard	.95	.95	Z-value	2-Sided
		Error	Lower	Upper		p-value
Intercept	.68	.15	.38	.98	4.45	.00
Participant (k = 13)	.00	.00	.00	.00	.40	.69
Q (I) = .16, <i>p</i> = .69						
Intercept	−197.52	108.58	−410.33	15.29	−1.82	.07
Year (k = 19)	.10	.05	−.01	.20	1.83	.07
Q (I) = 3.33, <i>p</i> = .07						
Intercept	.41	.11	.19	.63	3.61	.00
n in FOM (k = 19)	.01	.00	.00	.01	2.44	.01
Q (I) = 5.95, <i>p</i> = .01						

Discussion

The findings indicate a moderate overall effect size ($ES = .67$, 95% CI [.55–.78]), suggesting that use of AI technologies in education, on average, make a considerable contribution to student learning. This finding is consistent with previous meta-analyses (Chen et al., 2025; Tlili et al., 2025; X. Wang et al., 2024; Wu & Yu, 2024; Yoon, 2023).

[illegible]

as well as other systematic reviews (Almasri, 2024; Bond et al., 2024), clearly demonstrating a growing tendency of consensus that AI has the potential to foster educational outcomes, although the effectiveness can vary. To put it differently, the present study demonstrated that there were no statistically significant differences in effect sizes when analyzed by type of AI, learning outcome types, and quality levels of the studies. From these findings, considering the effect of the type of AI, one can argue that different types of AI (i.e., multi-type, chatbots, and AI-enabled assessment tools) are comparably and similarly effective on learning outcomes, and this finding is consistent with other meta-synthesis studies on AI use in education (Dong et al., 2025; Hwang, 2022; Zheng et al., 2023). Conversely, in their meta-analysis, Yi et al. (2025) discovered that the type of AI tools significantly impacts learning, with intelligent tutoring systems and adaptive learning systems showing notable effects, while pedagogical agents did not. However, they focused exclusively on mathematics education for K-12 students. In fact, a similar study on AI usage in mathematics education with elementary students revealed a non-significant influence of AI type on math learning (Hwang, 2022). From this finding, one can argue that how well AI is integrated might be more important than what kinds of AI tools are employed.

Similarly, when considering the influence of AI on learning outcomes, it was found that AI-enhanced interventions had comparable effects on academic achievement or higher-order thinking skills, with no significant differences observed between the two. However, one should interpret this finding cautiously for various reasons. For instance, in the current second-order meta-analysis, most of the first meta-analyses addressed the type of learning outcomes in a combined or mixed manner, while a few differentiated the learning outcomes as academic achievement or higher-order skills. Furthermore, only one FOM study addressed higher order skills explicitly. Nevertheless, numerous FOMs indicated that AI-based interventions not only positively influence academic achievement but also support higher-order thinking skills such as problem solving, project-based learning, and creative thinking or critical thinking, thereby spanning various levels of Bloom's taxonomy, albeit the effects varied based on how AI is used (subject domain, type of AI, or purpose of usage) (Ilgun Dibek et al., 2025; Sun & Zhou, 2024; J. Wang & Fan, 2025; Zheng et al., 2023). This finding aligns well with contemporary learning theories, such as constructivism, which emphasize the importance of active learning and self-regulation. Responsive AI systems can foster higher-order skills by providing scaffolding and guidance, supporting student autonomy, and enabling deeper learning outcomes (Guan et al., 2024; Hwang, 2022; Ilgun Dibek et al., 2025).

On the other hand, it is important to consider the risks associated with dependency on AI, as research indicates that overreliance on AI can negatively impact learning outcomes. When students excessively depend on AI, it can hinder their creative thinking, problem-solving abilities, and overall performance (Ilgun Dibek et al., 2025; Wang et al., 2024; Wang & Fan, 2025). Considering the effect size comparison based on the study quality, a non-significant result was observed. This suggests that the overall effect of AI is robust and is not necessarily significantly affected by the inclusion of studies having different qualities.

Considering educational level, it was identified as a significant moderator of AI's impact on learning. In the K12 and higher education context, AI had a stronger effect compared to the K12 context. This finding might indicate that AI-based interventions might be more successful when incorporated into mixed or higher-education settings relative to K12 alone. Although this finding is consistent with a significant portion of the existing literature (Deng et al., 2025; Wu & Yu, 2024; Yi et al., 2025; Zheng et al., 2023), caution is needed. Most studies have primarily explored AI implementation in higher or multiple educational settings, while there is comparatively little research focused on K-12 contexts, because while research on K-12 focuses on AI literacy courses, studies on the higher education context address the effect of AI incorporation into pedagogical methodologies. Furthermore, the availability of resources, opportunities, and various government funding may have significantly contributed to the adoption of AI within academic institutions (Hwang, 2022; Setiawan et al., 2025; Tili et al., 2025). The integration of AI in K-12 education might have been relatively ineffective due to several reasons, including an intense curriculum, inconsistencies between the curriculum and the AI applications, and the fact that not all AI products suit every educational need. Furthermore, learning and implementing AI tools, combined with existing workload demands, makes it challenging for teachers to find time for new technologies (Filiz et al., 2025). In the context of K-12 education, where students typically have lower cognitive capacity compared to higher education students, the complexity of AI tools and their outputs can lead to cognitive overload, potentially diminishing their practical utility in the classroom (Dong et al., 2025).

Another significant moderator impacting AI's influence was the education field, with computational science showing the strongest effect size, followed by language, multifield studies, and mathematics. The large effect in computational science aligns with Tili's (2024) findings, which emphasize the strong connection between AI principles and the nature of the computer science. AI tools, especially intelligent tutors, offer various affordances, such as code predictions, recommendations, and support for algorithmic thinking (Tili, 2024), making it a practical tool to adopt. Similarly, in the language field, AI-enabled customized tools can serve as valuable solutions by providing effective feedback and assessment mechanisms that enhance language skills, including reading, writing, and speaking (Chen et al., 2025). The current study revealed a moderate effect of AI use on student outcomes overall. However, its impact on student outcomes in the domain of mathematics appears limited, with some studies reporting only weak effects (Hwang, 2022), while others have found no significant influence (Steenbergen-Hu & Cooper, 2013). The variation in the types of AI analyzed or the target sample may have affected the results, and AI may not be as effective as expected in the domain of mathematics compared to other fields (Dong et al., 2025; Yi et al., 2025). This may also be related to the nature of mathematics itself, which is multidimensional and requires both abstract and concrete thinking. In other words, mathematical thinking involves not only applying rules to solve a given problem, but also understanding what to do and why (Young, 2017). In its current form, AI may not yet be fully capable of responding creatively to these complex learning needs. In sum,

studies revealed the potential of AI in education. On the other hand, teachers' readiness for AI and their attitudes towards it emerge as significant factors influencing its integration (Almasri, 2024; Baker et al., 2019). Furthermore, it is considered that teachers' concerns regarding AI's potential negative impact on student development and creativity could also impede its integration due to students' overreliance on AI (Bozkurt et al., 2023; Deng et al., 2025; Ilgun Dibek et al., 2025; Filiz et al., 2025).

The findings of the meta-regression suggest that the overall effect of AI on student outcomes is not influenced by sample size or the publication year of the first-order meta-analyses. This indicates stability of effects across different study scales and time periods. Conversely, the observed positive correlation between the number of primary studies and effect sizes suggests that meta-analyses incorporating a greater number of primary studies tend to report more substantial effects. This phenomenon may be indicative of enhanced statistical power, more comprehensive coverage of AI interventions, or the cumulative impact of diverse evidence on observed outcomes. This finding underscores the importance of including a sufficient number of primary studies when estimating AI's effectiveness in education.

Limitations and Further Research

The present study is constrained to the impact of AI applications on students' learning outcomes. However, it does not address the impact of AI applications on the behavior and attitudes of teaching staff/teachers. A compelling argument can be made for the necessity of research that delves into the ramifications of these policies on teaching staff, teachers, and school management. Additionally, the current study focuses solely on FOM studies published in English from 2020 to 2025. More extensive studies that incorporate FOM in English and other languages could also be considered. This finding suggests that language publication bias should be considered when interpreting the findings. It is evident that the effect sizes of certain moderator variables in this study do not demonstrate a balanced distribution. For instance, the educational domains are represented by two effect sizes each in the mathematics and computer science subgroups. This finding suggests that the observed differences in effect sizes across educational fields should be interpreted with caution. Another limitation of this review is the inclusion of three FOMs that did not provide lists of primary studies, which raises the possibility of overlap between the studies. Although a sensitivity analysis was conducted to assess the robustness of the results, potential duplication cannot be entirely excluded. This aspect should be taken into account when interpreting the findings.

Conclusion and Implications

Research on AI in education has expanded rapidly with the rise of generative platforms such as ChatGPT, prompting numerous studies on its impact on student outcomes, including academic achievement and higher-order thinking skills. However, findings remain mixed, limiting firm conclusions about AI's effectiveness (Dong et al., 2025; Sun & Zhou, 2024). While several meta-analyses have attempted to clarify this

relationship, their results are also inconsistent, highlighting the need for a second-order meta-analysis. Accordingly, this study synthesized evidence from prior meta-analyses to provide a comprehensive assessment of AI's effects on academic achievement and higher-order thinking skills, considering key moderator variables.

This second-order meta-analysis demonstrated a moderate overall effect size, indicating that AI applications and platforms contribute meaningfully to students' academic achievement and higher-order thinking skills. This finding underscores the potential of AI to enhance academic performance and higher-order skills when seamlessly integrated into pedagogical practices. However, it is imperative to exercise caution against overreliance, as this may impede creativity and independent problem-solving skills. The results of the study indicate the significance of allocating resources to investments in AI infrastructure, teacher training, and ensuring equitable access. To promote digital equity, policymakers and institutions should ensure that AI-supported learning tools are accessible across all educational levels and fields, preventing disparities due to technological or resource limitations. These measures can help maximize the benefits of AI for both academic achievement and higher-order thinking skills, particularly in under-resourced settings. These investments are crucial for ensuring that the advantages of AI-supported learning are disseminated across various educational levels and contexts. The substantial overall effect demonstrates the value of AI as an educational tool but also signals the need for more nuanced inquiry into long-term impacts, disciplinary differences, and the mechanisms through which AI fosters higher-order thinking.

The findings were consistent across different types of AI tools, learning outcomes, and study quality levels, suggesting that the manner in which AI is integrated into pedagogy may be more significant than the specific tool employed. Moderator analyses revealed that AI had stronger effects in higher education and mixed-level contexts compared to K-12 settings, and that disciplinary differences were evident, with computational sciences and language learning benefiting more than mathematics. The augmented efficacy of AI in higher education and mixed-level contexts suggests that its integration may be more effective in settings where technological infrastructure, pedagogical readiness, and curricular flexibility are more pronounced. This finding underscores the importance of using AI in line with students' developmental level and institutional capacity, while also adapting strategies to enhance the benefits of AI in K-12 settings. It also reveals the necessity for the allocation of supplementary resources, training, and support for early-level education to mitigate the discrepancy in AI effectiveness across diverse contexts. The disciplinary differences indicate that the fields of computational sciences and language learning are particularly well-suited to AI integration. In contrast, mathematics requires more targeted and domain-specific approaches. Researchers in this field must develop and evaluate AI applications tailored to mathematical reasoning and problem-solving processes.

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No information was directly collected from human participants.

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Data Availability Statement

Data will be available on request.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used ChatGPT and DeepL to enhance the language quality of the article. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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